

Efficiency of Iteratively-Converging Algorithms

Design and Analysis of Algorithms

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Iteratively-Converging Algorithms

- Given an increasing function $f(x)$, find a value of x in the range $[a, b]$ for which $f(x) = y$.
- For example, to compute $x = \sqrt{2}$, let $f(x) = x^2$ and solve $f(x) = 2$.
- How do we characterize the running time here?

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- How do we characterize the running time here?
- Two common alternatives:
 - Convergence rate: each step reduces additive error (at most $b - a$) by a factor of 2, giving $O(1)$ extra digits of precision in our answer x at each step (this rate, perhaps confusingly, is called “linear” convergence).
 - Steps needed to reduce error to ε factor of its initial value: $O(\log(1/\varepsilon))$ in this case, if we again use $b - a$ as a bound on additive error.

Applied to Optimization

- Suppose we want to find x^* minimizing $f(x)$.
- Error in current guess x measured with either $\|x - x^*\|$ or $f(x) - f(x^*)$.
- Same two common alternatives for measuring efficiency:
 - Convergence rate: characterizes the speed at which error decreases with time t . Example: linear convergence, where $f(x) - f(x^*)$ decreases exponentially with t , so we add $O(1)$ digits of accuracy in $f(x)$ at each step.
 - Steps needed to reduce error to ε factor of its initial value: E.g., $O(\log(1/\varepsilon))$ steps for linear convergence.